Nonword Humor

1. Nonword and single-word humor

Paper	What it does	What it shows	Link
Westbury, Shaoul, Moroschan & Ramscar (2016), *J. Memory and Language* 86	968 students rated 5,928 LINGUA nonwords; two experiments on 200 seven-letter nonwords (forced choice, rating)	Funniness ratings are consistent across people; lower unigram entropy predicts higher funniness (r ≈ 0.83 with model estimates); rude-sounding nonwords confound; Seuss words are low-entropy	DOI
Engelthaler & Hills (2018), *Behavior Research Methods* 50(3)	Crowdsourced 1–5 humor ratings for 4,997 English words from 821 people, by gender, age, education	Single words have reliable humor ratings, poorly predicted by valence, arousal or frequency; top: booty, tit, booby, hooter, nitwit	(open) PMC · data
Westbury & Hollis (2019), *JEP: General*; CogSci 2019	Models the Engelthaler norms with ~20 semantic and form features; predicts ~45,000 words	Beats the norms' split-half reliability; rarer words and improbable letters/phonemes (k, y, /u/) are funnier; sex, bodily, insult, animal meanings cluster high	(open) eScholarship
Lipps (1898), *Komik und Humor*	Philosophical treatise on comedy (German)	First explicit claim that nonwords are funny because we expect meaning and get none	(open) Gutenberg

Paper	What it does	What it shows	Link
Lubrich, Menninghaus et al. (2014), *Psych. of Aesthetics, Creativity, and the Arts* 8(1)	2×2 design varying rhyme and meter in humorous couplets	Rhyme and meter alone raise funniness and speed processing	DOI
Kentner et al. (2022), *Language and Cognition* 14(3)	German speakers rated and recalled reduplicative nonwords	Reduplication patterns shift affect, aesthetics and memory; ties to doubled-letter humor	(open) DOI
Palmann & Miller (2025), *Humor* 38(4)	Four phonological distance measures on a large rated pun dataset	Closer-sounding pun–target pairs are funnier on 2 of 4 measures; semantic distance not predictive	(open) PMC

2. Affect and meaning in nonwords

Paper	What it does	What it shows	Link
Sabbatino, Troiano, Schweitzer & Klinger (2022), WASSA @ ACL	Best-worst scaling of six emotions on 272 nonsense words (+68 real); character and phoneme regressors	Phonemes shift emotion (initial /s/ → joy); raters lean on similarity to real words; real-word model transfers weakly (r ≈ 0.17)	(open) ACL · arXiv
Reilly, Westbury, Kean & Peelle (2012), *PLoS ONE* 7	Tests whether form alone signals concrete vs. abstract	People have consistent form-based intuitions about nonword concreteness	(open) DOI
Westbury (2005), *Brain and Language* 93	Interference task: letter strings in round or spiky frames	Sound symbolism affects lexical decision implicitly	Scholar

Paper	What it does	What it shows	Link
Adelman, Estes & Cossu (2018), *Cognition* 175	Cross-linguistic phoneme–valence analysis	Phonemes predict word valence across languages	DOI
Needle, Pierrehumbert & Hay (2022), Cambridge UP chapter	Large pseudoword set varying length and phonotactic probability, rated for acceptability	Phonotactic likelihood and an apparent morphological parse predict acceptability; a wordlikeness covariate	Scholar

3. Nonwords and language models

Paper	What it does	What it shows	Link
de Varda, Gatti, Marelli & Günther (2024), *Computational Linguistics* 50(4)	18 language models define words and pseudowords; compared with human definitions	Pseudoword definitions are systematically tied to form; models capture pseudoword meaning	(open) ACL
Waal & Cassani (2025), Findings of ACL	Probes static and contextual LLM embeddings for pseudoword valence	Valence is encoded alone and in context; some models treat pseudowords like words in sentences	(open) ACL
Zangari, Marcuzzo et al. (2025), EMNLP	Swaps pun words for nonsense or non-punning words	Accuracy drops sharply; models stay overconfident that text is funny	Cardiff summary
Horvitz, Chen et al. (2024), ACL	LLMs edit jokes to remove humor, creating matched pairs	LLMs "unfun" well; synthetic pairs help train humor classifiers	Scholar

Paper	What it does	What it shows	Link
Yuan, Qu, Tawfelis & Färber (2025), Findings of IJCNLP-AACL	Pseudoword round/spiky and valence tasks in Dutch, Chinese and bilingual prompts	Prompt language changes model behavior; multilingual models mimic language-specific patterns	(open) PDF

4. Sound symbolism in VLMs and LLMs

Paper	What it does	What it shows	Link
Alper & Averbuch-Elor (2023), NeurIPS	Zero-shot probing of CLIP and Stable Diffusion with 648 constructed pseudowords on a sharp–round axis; user study	Both encode sound symbolism (AUC 0.77 CLIP, 0.74 SD); humans recover pseudowords from images above chance; text-only encoders mixed	(open) arXiv · project
Loakman, Li & Lin (2024), EMNLP	Kiki–bouba shape, mil–mal size and iconicity rating tasks on VLMs and LLMs	Size symbolism is easier than shape; iconicity agreement grows with model size	(open) arXiv
Verhoef et al. (2024)	Compares four VLMs on bouba–kiki	Cited as finding CLIP most human-like; verify citation before use	Scholar
Iida & Funakura (2024), CogSci	Japanese text and vision-language models on shape symbolism	Results diverge from Alper & Averbuch-Elor; effect may be language-specific	(open) eScholarship
Revisiting bouba-kiki in CLIP (2025), arXiv:2507.10013	Tests CLIP ResNet and ViT on abstract shapes and novel words	Reviews mixed prior results; checks robustness of CLIP's associations	(open) arXiv

Paper	What it does	What it shows	Link
Milička et al. (2025), arXiv:2501.05643	LLMs generate Czech and German iconic pseudowords; LLM participants guess meanings	LLMs both produce and interpret sound-symbolic pseudowords	(open) arXiv
Phoneme sound symbolism in LLMs (2026), arXiv:2603.17306	GPT-4o, Claude Sonnet 4, Gemini 2.5 Flash rate minimal-pair pseudowords on nine dimensions	Models agree on consistent multidimensional phoneme meanings from text alone	(open) arXiv

5. Human sound symbolism (background)

Paper	What it shows	Link
Köhler (1929/1947), *Gestalt Psychology*	Original maluma/takete observation	Scholar
Sapir (1929), *JEP* 12	~80% matched "mal" to a large table, "mil" to a small one	DOI
Ramachandran & Hubbard (2001), *J. Consciousness Studies* 8	Named kiki/bouba; cross-modal mapping account	Scholar
McCormick et al. (2015), CogSci	Roundness/pointedness ratings for many pseudowords; sharp-to-round phoneme ranking used by Alper & Averbuch-Elor	(open) CORE
Fort & Schwartz (2022), *Scientific Reports* 12	Effect grounded in spectral balance and temporal continuity	(open) DOI
Ćwiek et al. (2022), *Phil. Trans. R. Soc. B* 377	Effect holds across 25 languages and writing systems	(open) DOI
Blasi et al. (2016), *PNAS* 113	Sound–meaning biases across thousands of languages	(open) DOI
Lacey et al. (2020), *Cognitive Science* 44	Acoustic parameters drive sound-to-shape mapping (RSA)	DOI

Paper	What it shows	Link
Sidhu et al. (2021), *Psychonomic Bulletin & Review* 28	Maluma/takete effect in real English nouns	Scholar
Monaghan et al. (2014), *Phil. Trans. R. Soc. B* 369	Corpus-level systematicity in English	(open) DOI
Winter & Perlman (2021), *Glossa* 6	Size sound symbolism in the English lexicon	Scholar
Maurer, Pathman & Mondloch (2006), *Developmental Science* 9	Toddlers show the bouba–kiki effect	Scholar
Ozturk, Krehm & Vouloumanos (2013), *J. Exp. Child Psych.* 114	Sound–shape sensitivity at 4 months	Scholar

6. Humor theory

Paper	What it shows	Link
Schopenhauer (1818), *The World as Will and Representation*	Incongruity theory: mismatch between concept and object	(open) Gutenberg
Hurley, Dennett & Adams (2011), *Inside Jokes*	Humor rewards detecting false beliefs ("cognitive debugging")	MIT Press
Suls (1972)	Two-stage model: incongruity, then resolution	Scholar
Attardo & Raskin (1991), *Humor* 4	General Theory of Verbal Humor	Scholar
Goel & Dolan (2001), *Nature Neuroscience* 4	Pun and semantic jokes use dissociable brain regions	DOI
Mobbs et al. (2003), *Neuron* 40	Humor activates the mesolimbic reward system	Scholar
Gervais & Wilson (2005), *Quarterly Review of Biology* 80	Evolutionary account of laughter and humor	Scholar

7. Methods and tools

Paper	What it shows	Link
Westbury, Hollis & Shaoul (2007), *The Mental Lexicon* 2	LINGUA: the Markov n-gram nonword generator used in Westbury et al. (2016)	Scholar
Shannon (1948), *Bell System Technical Journal* 27	Defines entropy and surprisal, the basis of the humor measure	Scholar
Binder, Medler, Westbury et al. (2006), *NeuroImage* 33	Left fusiform gyrus responds in graded steps to orthographic wordlikeness	Scholar
Baayen (2010), *The Mental Lexicon* 5	Contextual measures explain word frequency effects; why Westbury uses information theory over raw frequency	Scholar